Electrical properties of Molecules

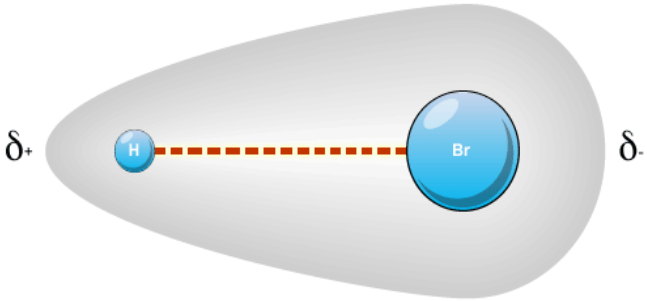
What are polar bonds and polar molecules?

Polar and Nonpolar Bonds

- There are two types of covalent bonds
  - Nonpolar Covalent Bonds (equal share of electrons)
  - Polar Covalent Bonds (unequal share of electrons)

Polar Covalent Bond

- A Polar Covalent Bond is *unequal* sharing of electrons between two atoms (H-Cl) due to differences in electronegativity.
- In a polar covalent bond, one atom typically has a negative charge, and the other atom has a positive charge



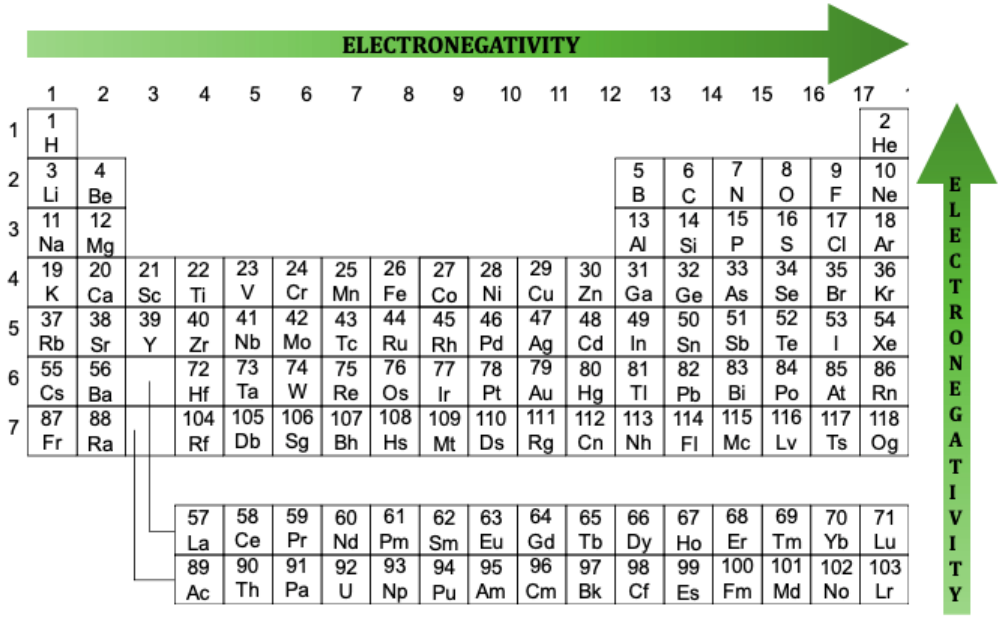
Nonpolar Covalent Bond

- A Nonpolar Covalent Bond is an *equal* sharing of electrons between two atoms (Cl-Cl, N-N, O-O) with similar electronegativity.

Bond polarity & electronegativity

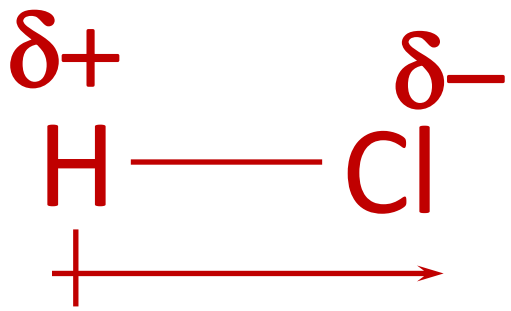
"Electronegativity is the ability of an atom in a molecule to attract the molecule's electrons toward itself."

Electronegativity is used to determine polarity of a bond between two atoms.



How to show a bond is polar

- Isn't a whole charge just a partial charge
- δ+ means a partially positive (less electronegative)
- δ– means a partially negative (more electronegative)



- The Cl pulls harder on the electrons
- The electrons spend more time near the Cl

Polar Molecule

A polar molecule, is a molecule that has a charge on each side of the molecule, that is not cancelled out. It has a region of partial charge. One end is slightly positive one end is slightly negative. They are generally asymmetrical, with an uneven distribution of the electrons.

A molecule with two poles is called a **dipole**. When you measure the amount of polarity of a molecule, the result is called the **dipole moment**.

Examples of Polar Molecules

HCl, CH_3Cl, NH_3, H_2O, C_2H_5OH, SO_2 etc.

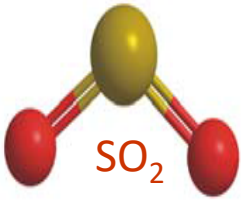
Non polar Molecule

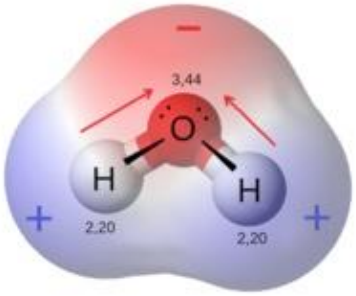
If a molecule is non-polar, then the atoms either share the electrons evenly, e.g. they have a non-polar bond, or the polar bonds are symmetric, in the cases of carbon dioxide or carbon tetrachloride. In those molecules, there are dipoles but they cancel out due to the symmetry.

Examples of Nonpolar Molecules

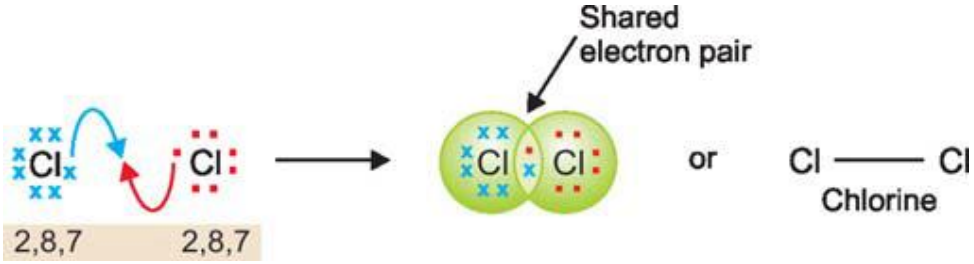
H_2, Cl_2, CH_4, CO_2, CCl_4, C_6H_6, etc.

Polar molecules





Nonpolar molecules

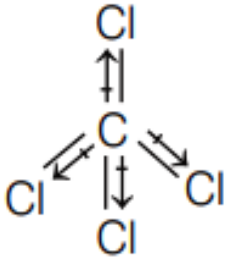


Symmetrical Molecules

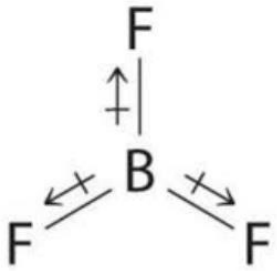
- Because of symmetry, molecules that have polar bonds are overall a **nonpolar molecules** (+ and – charges cancel out or balance out)

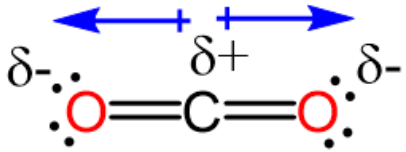
Examples:

CCl_4



BF_3





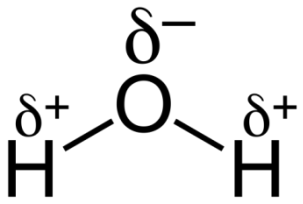
CO_2

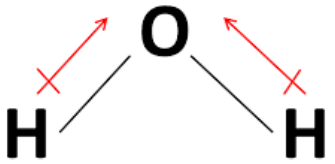
Asymmetrical Molecules

- If a molecule has polar bonds (and there is **no symmetry** to cancel out + and – charges), the molecule is polar.

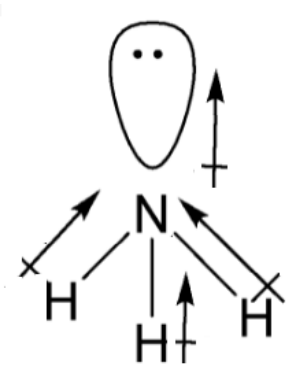
Examples:

H_2O

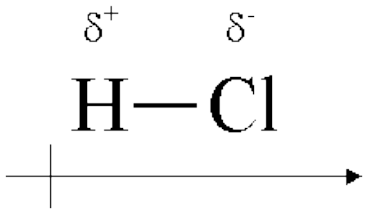


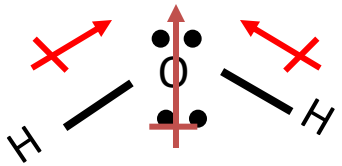


NH_3

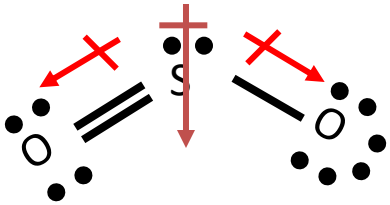


HCl

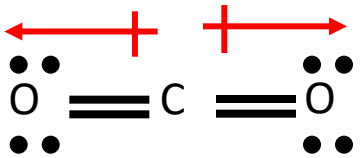




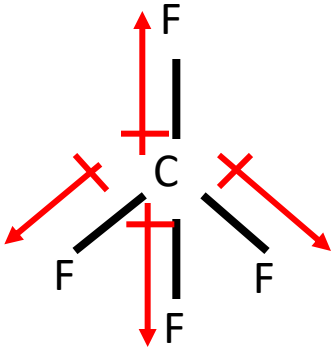
dipole moment
polar molecule



dipole moment
polar molecule



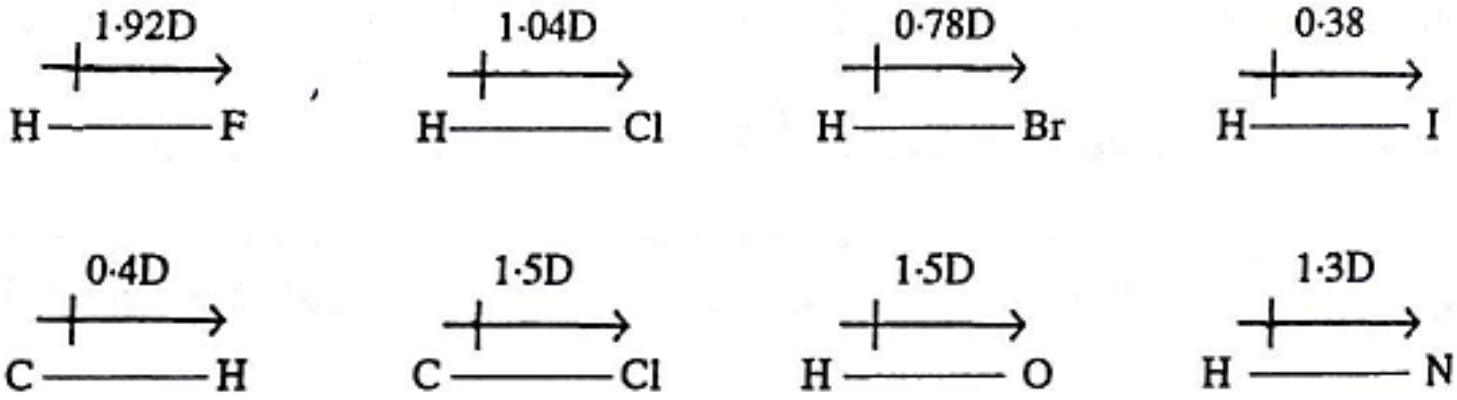
no dipole moment
nonpolar molecule



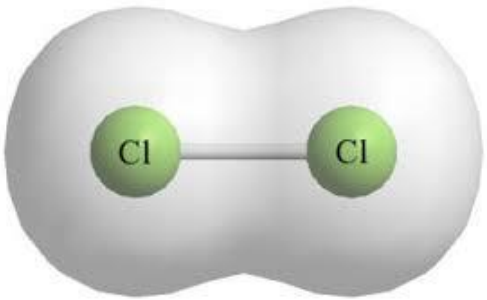
no dipole moment
nonpolar molecule

Bond Moments

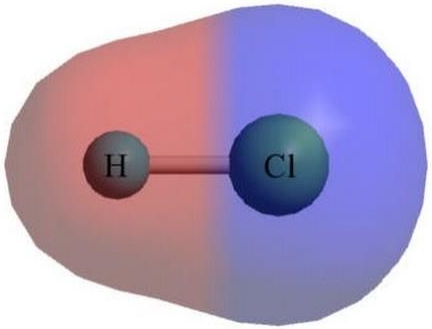
Any bond which haws a degree of polarity has a dipole moment. This is called **bond moment**.



In a diatomic molecule, the bond moment corresponds to the dipole moment of the molecule.



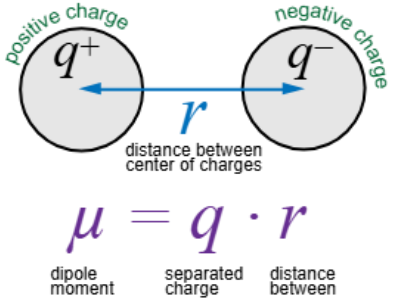
μ = 0, Non-polar



μ = 1.04D, Polar

Dipole moment

A dipole moment(μ) is a measure of the polarity of a molecule, indicating the separation of positive and negative charges and is a vector quantity with magnitude and direction, which is the product of the charge q and the distance r between the charges.



$$\mu = q \times r$$

To maintain electrical neutrality, the charges on both ends of an electrically neutral diatomic molecule must be equal in magnitude and opposite in sign. However, in above Equation, q refers only to the magnitude of the charge and not to its sign, so μ is always positive.

Dipole moments are usually expressed in debye units (D), named for Peter Debye. The conversion factor is

$$1D = 3.336 \times 10^{-30}\ Cm$$

Units of dipole moments:

If an electron is placed 1 Å from an equal positive charge,

$$\mu = q \times r$$

$$\mu = 4.8 \times 10^{-10} \text{ esu} \times 10^{-8} \text{ cm}$$

$$= 4.8 \times 10^{-18} \text{ esu-cm}$$

The quantity 10^{-18} esu-cm is called one **debye** (D). 1D = 10^{-18} esu-cm;

So, $\mu = 4.8$ D

In SI system

Dipole moment,

$$\mu = 1.602 \times 10^{-19} \text{ C x } 10^{-10} \text{ m}$$

$$= 1.602 \times 10^{-29} \text{ C m}$$

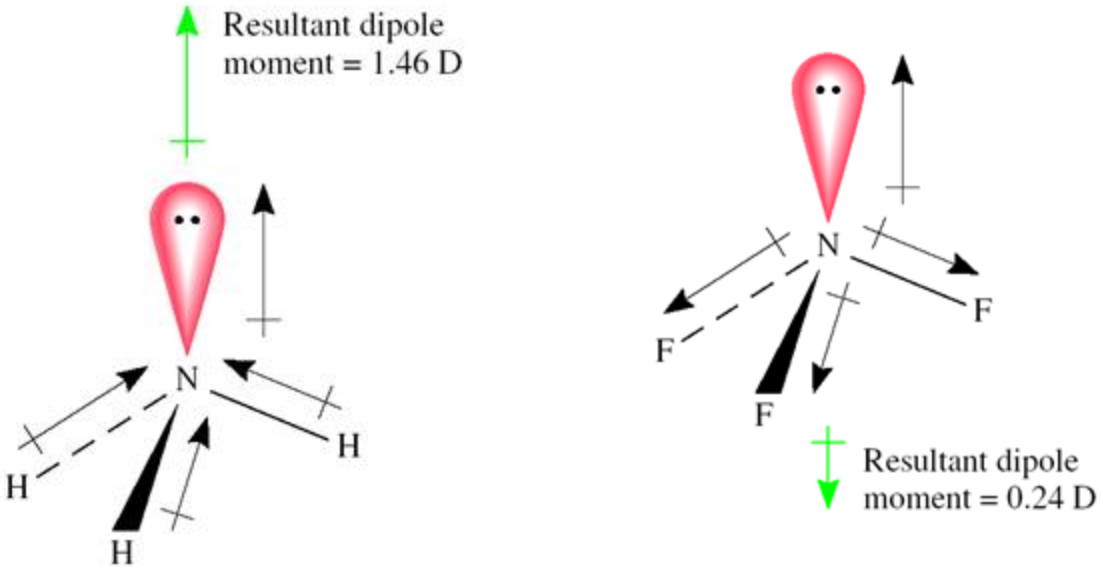
Here, 1D = 3.336×10^{-30} C m

So, $\mu = (1.602 \times 10^{-29} \text{ C m}) \times \frac{1}{3.336 \times 10^{-30} Cm} = 4.8D$

Dipole moment is a **vector quantity**

Molecule containing three or more atoms, the net dipole moment is the **vector resultant** of the individual bond moments.

Bond moments and resultant dipole moments in NH_3 and NF_3.



Polarizability

When an electric field is applied, the negatively charged electrons and positively charged nuclei within an atom or molecule experience opposing forces, causing a separation of charge and the formation of an induced dipole moment.

Polarizability refers to the tendency of an atom or molecule's electron cloud to distort or shift under the influence of an electric field, leading to the formation of temporary dipoles.

Polarizability is the measure of how easily an electron cloud can be distorted by an external electric field.

As electric field has induced a dipole moment,

$$\mu_{ind} \propto E_i$$

where, μ_{ind} is induced dipole moment and E_i is strength of electric field

Or,

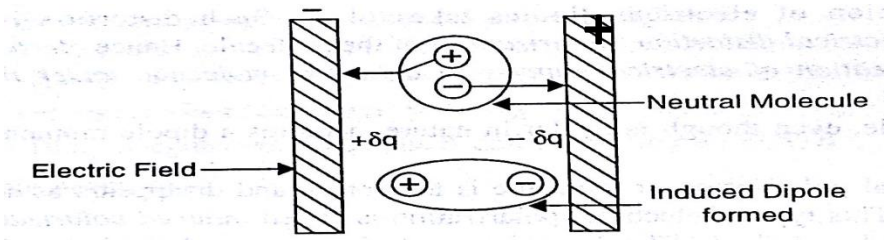
$$\mu_{ind} = \alpha E_i$$

where, α is **polarizability**

If E_i = unit electric field strength, then $\mu_i = \alpha$

Induced or Distortion Polarization

The creation of positive or negative charges in a neutral molecule under the influence of an electrical field is known as **electrical polarization** of the molecule.



It is temporary, disappears as the electric field is removed

- This type of polarization known as **induced polarization**
- dipole formed is called **induced dipole**

Types of induced polarization

Two types:

a) Electric polarization, $\mathbf{P_e}$: electron will be distorted with respect to the nuclei towards the positive plates.

b) Atomic polarization, $\mathbf{P_a}$: the nuclei will be distorted with respect to each other.

Induced polarization = Atomic polarization + Electric polarization

$$Pi = Pa + Pe$$

According to Clausius and Mosotti relation

$$4/3\pi N\alpha = (D-1)/(D+2)\ M/\rho$$

Where,

M= Molecular weight of substance

ρ = Density of substance

N= Avogadro's number

α = Distortion polarizability

D = Dielectric constant of medium

Left side of the above equation is called molar polarization of substance and is denoted By P_M. Hence

$$P_M = 4/3\pi N\alpha = (D-1)/(D+2)\ M/\rho$$

In non polar molecules, the polarization is all induced. Hence P also stand for induced Polarization Pi in case of nonpolar molecules only. So

$$Pi = P_M = 4/3\pi N\alpha = (D-1)/(D+2)\ M/\rho$$

Orientation Polarization

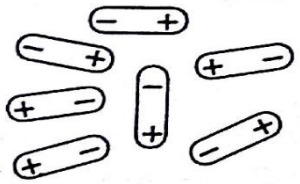
When polar molecules are placed in an electric field, two effects will arise:

Induced polarization (P_i):

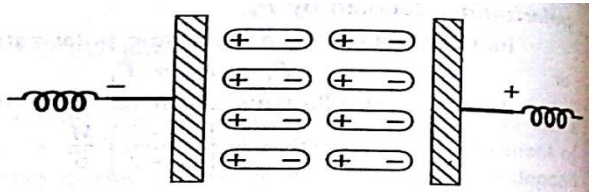
Usual distortion of positive and negative charges

Orientation polarization (P_o):

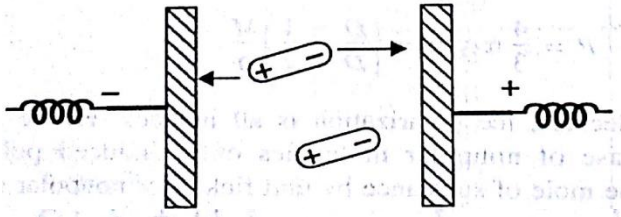
Molecules occupy some mean position between the direction of electric field and the their original position



Polar molecules are randomly oriented in all direc tions, in absence of electric field.



If the polar molecules are stationary, they would orient at 180° to the direction of electric field (*i.e.*, perpendicular to the electric plates).



. Due to thermal agitation, molecules occupy mean position between the direction of electric field and original positions of molecules, it is orientation polarization.

For a polar molecule

Molar polarization = Induced polarization + Orientation polarization

$$P_M = P_i + P_o \quad \text{----------(1)}$$

But for dielectric constant measurments

$$P_M = (D-1)/(D+2)\ M/\rho \quad \text{----------(2)}$$

We know $P_i = 4/3\pi N\alpha$ ----------(3)

$$\therefore \quad P_M = (D-1)/(D+2)\ M/\rho = 4/3\pi N\alpha + P_o \quad \text{----------(4)}$$

But Debye has shown that

$$P_o = 4/3\pi N \,.\, \mu^2/3kT \quad \text{----------(5)}$$

From eq. (4) and eq.(5), we get

$$P_M = 4/3\pi N\alpha + 4/3\pi N \,.\, \mu^2/3kT \quad \text{----------(6)}$$

In eq. (6) the first term on R.H. side is constant (as N and α are constant) and μ is also constant in the second term. Hence eq. (6) may be express as

$$P_M = A + B/T$$

Induced polarization is

temperature independent

depends on the ratio of dielectric constant & density

Orientation polarization is

temperature dependent

Molar polarization of nonpolar molecules,

- Do not have permanent dipole
- Hence no orientation polarization
- Independent of temperature

Molar polarization of polar molecules

- Have permanent dipole
- Have orientation polarization
- Temperature dependent

Thus, one can distinguish between polar and nonpolar molecules

Polarization & Polarisability

Polarization is the process by which the electron cloud of an atom or molecule is distorted in the presence of an external electric field. This distortion leads to the separation of positive and negative charges within the molecule, creating a dipole moment.

Polarizability is a quantitative measure of how easily the electron cloud of an atom or molecule can be distorted by an external electric field. It is often denoted by the symbol α (alpha).

- Electrical field separates the centres of the positive and negative charges.
- Electric field has induced a dipole moment.

$$\mu_{ind} \propto E_i$$

where, μ_{ind} is induced dipole moment and E_i is strength of electric field

or

$$\mu_{ind} = \alpha E_i$$

where, α is polarizability

Clausius-Mosotti Equation

The relation between dielectric constant (ratio of the permittivity of the substance to the permittivity of the free space) and induced polarization was deduced by **Clausius–Mossotti** and generally known as the Clausius–Mossotti equation. When polar or non-polar molecule is placed in an electric field there will be small displacement of the electrical centers which will be induced in the molecules.

If μ_i is the electrical moment of the induced dipole produced by a field of intensity **F acting on a single molecule**, then

$$\mu_i = \alpha_D F \qquad (1)$$

The constant α_D is called the polarizability of the molecule

Suppose the strength of a uniform electric field produced by the two charged plates is E_0, then in any non-polar medium the field strength is reduced to E. It can be shown by electrostatics that

$$E_0 = E + 4\pi I \qquad (2)$$

or, $E_0 - E = 4\pi I$

or, $E(E_0/E - 1) = 4\pi I$

or, $E(\varepsilon - 1) = 4\pi I$ (3)

where ε is called the dielectric constant of the medium and I is the induced electric moment per unit volume, i.e.,

$$I = \mu_i n \quad (4)$$

where n is the number of molecules per mL

The electrical intensity F is made up of several parts

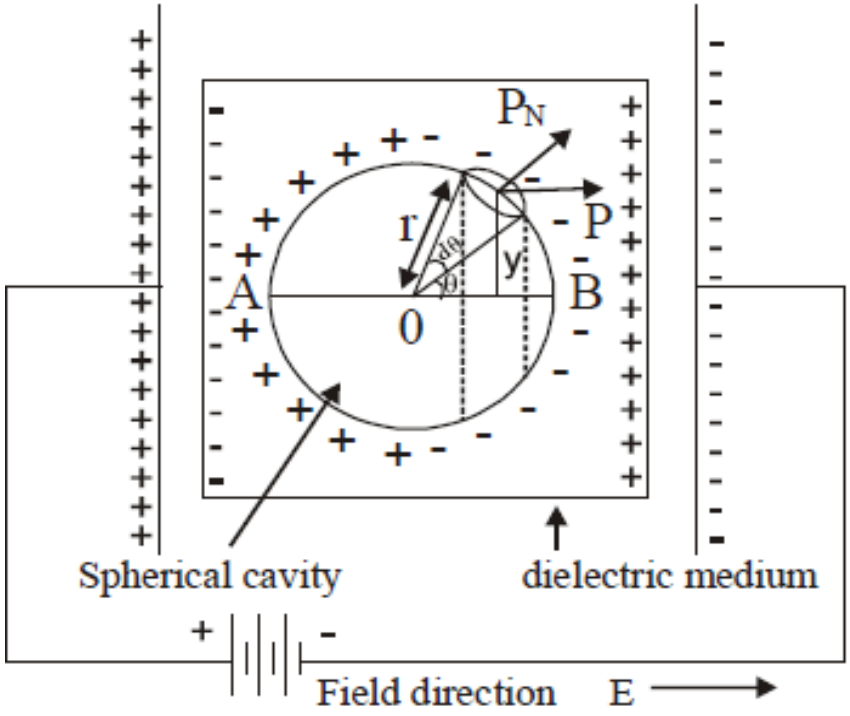
(i) Field charge on the plates, i.e., E_0

(ii) Force - $4\pi I$ due to the charge induced in the surfaces of the dielectric in contact with the plates.

(iii) $4/3\pi I$ resulting from the charge induced on the surface of the spherical cavity.

(iv) The filed caused by the molecule within the cavity.

In case of liquids and gases, the molecules are oriented in the absence of an external field. Therefore, the filed caused by the molecule within the cavity may be neglected.


P_N
P
r
$d\theta$
θ
y
A
B
0
Spherical cavity
dielectric medium
Field direction
E

Thus, the electric intensity F may be written as

$$F = E_0 + \frac{4}{3}\pi I - 4\pi I \quad (5)$$

Introducing eq.2 into eq.5, we get

$$F = E + 4\pi I + \frac{4}{3}\pi I - 4\pi I = E + \frac{4}{3}\pi I \quad (6)$$

Introducing eq.3 into eq.6, we get

$$F = E + \frac{1}{3}E(\varepsilon - 1) = E + \frac{E\varepsilon}{3} - \frac{E}{3}$$

$$F = \frac{2E}{3} + \frac{E\varepsilon}{3} = \frac{E}{3}(2 + \varepsilon) \quad (7)$$

Substituting eq.7 in eq.1 we get,

$$\mu_i = \alpha_D . \frac{E}{3}(2 + \varepsilon) \quad (8)$$

Substituting eq.8 into eq.4, we get

$$I = n\alpha_D \frac{E}{3}(2 + \varepsilon) \quad (9)$$

Substituting eq.9 into eq.3, we get

$$E(\varepsilon - 1) = 4\pi n\alpha_D \frac{E}{3}(2 + \varepsilon)$$

Or

$$\frac{\varepsilon - 1}{\varepsilon + 2} = \frac{4}{3}\pi n\alpha_D \quad (10)$$

If ρ is the density of the medium between the charged plates and M is the molecular Weight, then the number of molecules n in unit is Nρ/M, where N is the Avogadro's Number i.e,

$$n = \frac{N\rho}{M} \tag{11}$$

On substituting eq .11, into eq.10, we get

$$\frac{\varepsilon-1}{\varepsilon+2} \cdot \frac{M}{\rho} = \frac{4}{3}\,\pi N \alpha_D \tag{11A}$$

The left side of the equation is given by the symbol P_M and is called molar Polarization of the material, i.e,

$$P_M = \frac{\varepsilon-1}{\varepsilon+2} \cdot \frac{M}{\rho} = \frac{4}{3}\,\pi N \alpha_D \tag{11B}$$

Eq. (11B) is known as Clausius- Mosotti equation.

$$P_i = P_M = \frac{4}{3}\,\pi N \alpha_D \tag{12}$$

Debye Equation

In 1912 Debye established the relation between molar polarization and temperature and generally known as the Debye equation. When polar molecules are placed in an electric field the molecules orient themselves in the direction of the field.

The electric moment, μ_m per molecule is given by

$$\mu_m = \alpha F$$

$$\text{or, } \mu_m = (\alpha_D + \alpha_0)F$$

$$\text{or, } \mu_m = \alpha_D F + \alpha_0 F \qquad (12A)$$

$$\text{or, } \mu_m = \mu_i + \bar{m} \qquad (13)$$

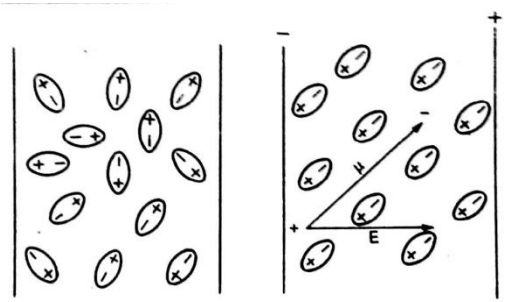


Fig. 5·6. Orientation of molecules : (a) in the absence of an electric field (b) in the presence of an electric field.

where μ_i the induced polarization, $\bar{m}$ is the moment due to orientation polarization.

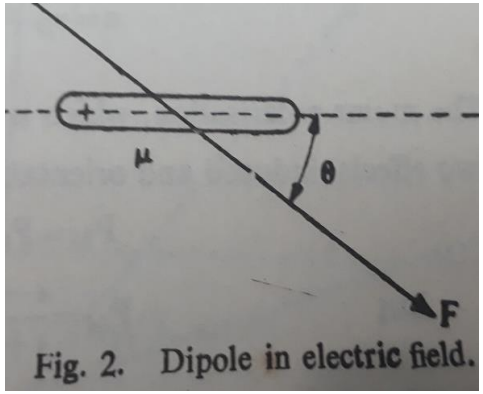


Fig. 2. Dipole in electric field.

If μ is the permanent molecular moment and molecular dipoles are oriented that their axis make an angle θ with the direction of the field, the potential energy of a molecule is given by

$$U = - \mu F \cos\theta \qquad (14)$$

Now according to Boltzmann's law, the number of molecules with their dipole axes in the direction with a solid angle (dw) is given by

$$dN = Ae^{-U/kT}dw$$

The total number of molecules can be obtained by integrating the above equation over the possible direction, i.e., for solid angle between 0 and 2л

thus, total no. of molecules = $\int_0^{2л} Ae^{-U/kT} dw$

$$= \int_0^{2л} Ae^{\mu F\cos\theta/kT} dw$$

and the total moment in the direction of the field is given by

$$total\ moment = \int_0^{2\pi} Ae^{\mu F\ cos\theta/kT} . \mu\ cos\theta\ \ dw$$

Therefore, the average moment per molecule $\overline{m}$ in the direction of the field

$$\bar{m} = \frac{\int_0^{2\pi} Ae^{\mu F\ cos\theta/kT} . \mu\ cos\theta\ \ dw}{\int_0^{2\pi} Ae^{\mu F\ cos\theta/kT} . dw}$$

Substituting μF/kT = x and cosθ = t and making use of the relation dw = 2π sinθ dθ
We get

$$\bar{m} = \frac{\mu \int te^{tx} dt}{e^{tx} dt}$$

$$\bar{m} = \frac{e^x + e^{-x}}{e^x - e^{-x}} - \frac{1}{x} = \left(\coth x - \frac{1}{x}\right) = L(x) \qquad (15)$$

Where L(x) is the Langevin function. If x is small particularly for small field strength, it is found that $L(x) \approx 1/3x$

and then equation (15) gives, $\overline{m} = 1/3\mu x = \mu^2/3kT \times F$ (16)

On comparing eq. 12A and 13, we get

$$\overline{m} = \alpha_0 F \quad (17)$$

On comparing eq. 16 and 17, we get

$$\alpha_0 F = \frac{\mu^2}{3kT} F \quad or\ \alpha_0 = \frac{\mu^2}{3kT} \quad (18)$$

But the total polarization, α is given by

$$\alpha = \alpha_D + \alpha_o = \alpha_D + \frac{\mu^2}{3kT} \quad (19)$$

The molar polarization, which is assume from dielectric constant $\frac{\varepsilon-1}{\varepsilon+2}.\frac{M}{\rho}$

Is the sum of the two effect, induced and orientation polarization i.e.,

$$P_M = P_i + P_o$$

But

$$P_M = \frac{\varepsilon - 1}{\varepsilon + 2}.\frac{M}{\rho} \quad (20)$$

If α represent the total polarisability, we can write

$$\frac{\varepsilon-1}{\varepsilon+2}\cdot\frac{M}{\rho}=\frac{4\pi N}{3}\alpha \qquad (21)$$

Substituting eq.19 in eq.21 we get

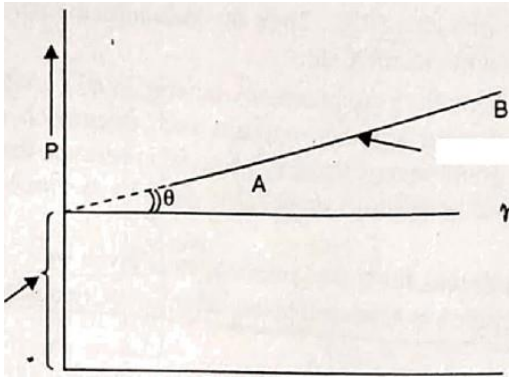
$$\frac{\varepsilon-1}{\varepsilon+2}\cdot\frac{M}{\rho}=\frac{4\pi N}{3}\left(\alpha_D+\frac{\mu^2}{3kT}\right) \qquad (22)$$

This equation 22 is known as the Debay equation. It is clear from the equation that

$$P_M = P_i + P_o$$

Such that $P_i=\frac{4\pi N\alpha_D}{3}$, the induced polarization and $P_o=\frac{4}{3}\pi N\cdot\frac{\mu^2}{3kT}$ Is the orientation polarization.

$$P_M = A + B/T$$



$$\frac{1}{T}$$

Frequency and temperature dependence of polarization

On the application of an alternating electric field, the polarization process occurs as a function of time. The polarization P(t) as a function of time ‘t’ is give by

$$P(t) = P\,[1 - e^{\left(\frac{-t}{t_r}\right)}]$$

Where,‘P’ is the maximum polarization attained on external applied static field and ‘t_r’ is the relaxation time for the particular polarization process. The relaxation time is a measure of the time scale of a polarization process. This varies for different polarization process.

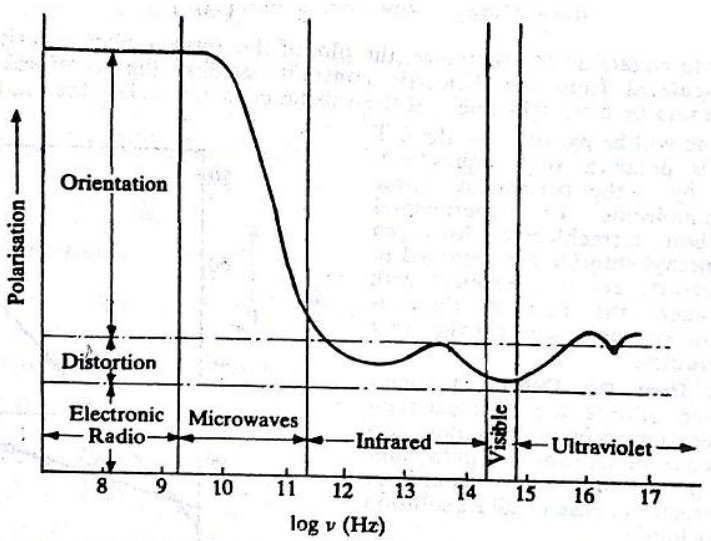


Fig. 1. The dependence of P_M on frequency.

Frequency dependent polarization

1. Electronic polarization is very fast. The frequency of the voltage is very high, which is in the optical range (≈1015 Hz). Thus it occurs at all frequencies.

2. Ionic polarization is slower than the electronic polarization, because the mass of the ions are heavier than the electrons cloud. It is occurs in the infra-red frequency region (1013 Hz).

3. Orientation polarization is slower than the ionic polarization and which is occurs only at electrical frequency (106 Hz).

4. Space charge polarization is the slowest process and occurs only at power frequencies (102 Hz).

5. Thus at lower frequencies, the value of the total polarization is very high and at higher frequencies (optical frequencies) the value of the total polarization is very small.

Determination of Dipole Moment

Methods available for the dipole moment measurements are:

1. Temperature Method
2. Ebert's Method
3. Refraction Method
4. Dilute Solution Method

Temperature Method

$$P_o = \frac{4}{3}\pi N \cdot \frac{\mu^2}{3kT}$$

We know that

$P = A + B/T$

Where P= Total molar pollarization

If total polarization P_1 and P_2 be determine at two temperatures T_1 and T_2, then

$P_1 = A + B/T_1$

and $P_2 = A + B/T_2$

$\therefore \quad P_1 - P_2 = B\,[\,1/T_1 - 1/T_2] = B\,[\,(T_2 - T_1)/T_1T_2]$

Or $B = (P_1 - P_2)\,[\,T_1T_2/(T_2 - T_1)]$ ----------(1)

Now $B = 4\pi N\mu^2/9k$

$\therefore \quad \mu = \sqrt{9kB/4\pi N}$

Putting the standard values of π , N and k, the equation becomes

$\mu = 0.0128 \times 10^{-18} \sqrt{B}$ esu-cm

$\mu = 0.0128 \times \sqrt{B}$ debey ----------(2)

Putting the value of B from equation (1) in equation (2), μ is claculated

Ebert's Method

The polar molecules in a gas state, when kept in electric field, show both induced and oriantation polarizations, but the same polar molecules in solid state, show only induced polarization. Hence for a given polar substance, total molar polarization is:

In gas state: $P_{gas} = P_i + P_o$ ---------- (1)

In solid state: $P_{solid} = P_i$ ----------(2)

Hence from equation (1) and (2)

$$P_{gas} - P_{solid} = P_o \text{ ----------(3)}$$

But $P_o = 4\pi N/3[\ \mu^2/3kT]$

$$P_{gas} - P_{solid} = 4\pi N/3[\ \mu^2/3kT]$$

Or $$\mu = \sqrt{9kT/4\pi N\ (P_{gas} - P_{solid})} \text{ --- (4)}$$

Thus μ is calculated with the help of equation (4)

Refraction Method

This method involves the measurement of dielectric constant of the vapor in terms of refractive index only at one temperature.

We have,

$$P = P_i + 4\pi N/3.\ \mu^2/3kT$$

Or $$\mu = \sqrt{(9k/4\pi N)} .\ \sqrt{\{(P-P_i).\ T\}} = 0.0128\sqrt{[(P-P_i)T]}$$

Where P_i is the distortion polarization. Thus, for the determination of μ, the total Polarization P, and the distortion polarization P_i must be known accurately.

Where , $P_i = 4/3\pi N\alpha$ can be calculated if α value is known.

We know,

$$P = \{(\varepsilon -1)/(\varepsilon+2)\}.\ M/d$$

Maxwell showed that for the same electromagnetic vibration, the refractive index (n) is related to the dielectric constant by the expression,

$$\varepsilon = n^2$$

So for non- polar molecules,

$$P = (n^2-1)/(n^2+2) \cdot M/d$$

Refractive index is measured with visible light that is at a lower wavelength but actually dielectric constant is measured at the higher wavelength that is why there was a need for some correction factor. The following formula is used to measure refractive index at a higher wavelength.

$$n = n_\infty + (a/\lambda^2)$$

Where a is a constant; n is the refractive index at wavelength λ, and n_∞ that for long wavelength.

Application of Dipole Moment

(i) The dipole moment helps to predict whether a molecule is polar or non-polar.

(ii) It helps to determine the shape of molecules

(iii) It helps to distinguish between cis- and trans-isomers

.

(iv) The percentage of ionic character can be calculated from dipole moment

(v) It helps to distinguish between ortho, meta and para-isomers.

(vi) Bond angle can be calculated with the help of it.

(vii) It helps to determine the resonance structure.

.

Application of Dipole Moment

- **Determining molecular polarity:**

A molecule with a significant dipole moment is considered polar, while a molecule with zero dipole moment is non-polar.

Compound	Dipole Moment (D)
HF	1.82
HCl	1.08
HBr	0.82
HI	0.44
O_2	0
CO_2	0

Polarity of a bond:

Dipole moment gives us an indication of the polar character of a bond.

Bond	Dipole moment (D)	Bond	Dipole moment (D)
H-F	1.9	C=O	2.3
H-Cl	1.0	C-F	1.4
H-Br	0.8	C-Cl	1.5
H-I	0.4	C-Br	1.4
H-O	1.5	C-I	1.2
H-N	1.5	C-O	0.7
C-H	0.4	C-N	0.2

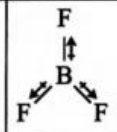
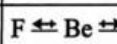
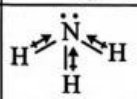
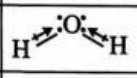
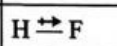
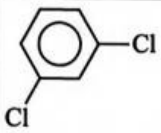
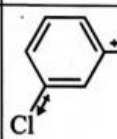
•Identifying molecular symmetry and shape

A molecule's shape can be determined by analyzing its dipole moment because a non-zero dipole moment indicates an asymmetrical distribution of electrons, which in turn points to a specific molecular geometry, meaning the molecule is not perfectly symmetrical; conversely, a zero dipole moment suggests a symmetrical structure where individual bond dipoles cancel each other out

Molecules with symmetrical structures have a zero dipole moment, whereas asymmetrical molecules exhibit a net dipole moment.

If $\mu = 0$ compound is nonpolar and symmetrical
Eg. CO_2, BF_3, CCl_4, CH_4, BeF_2 etc.

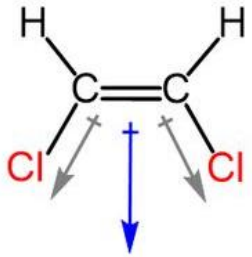
If $\mu \neq 0$ compound will be polar and unsymmetrical
H_2O, SO_2, NH_3, Cl_2O, CH_3Cl, $CHCl_3$ etc.

Species	Structure	Dipole moment
BF_3		$\mu = 0$
BeF_2		$\mu = 0$
CO_2		$\mu = 0$
		$\mu = 0$
NH_3		$\mu \neq 0$
H_2O		$\mu \neq 0$
HF		$\mu \neq 0$
		$\mu \neq 0$
NF_3		$\mu \neq 0$

•Distinguishing between isomers

Cis and trans isomers can be differentiated based on their dipole moment values, with the trans isomer usually having a higher dipole moment.

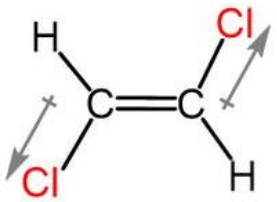
***cis*-1,2-Dichloroethene**



net dipole moment

μ = 1.9 D

***trans*-1,2-Dichloroethene**



Opposite dipoles cancel

μ = 0 D

•Calculating percentage ionic character

Dipole moment data can be used to estimate the percentage of ionic character in a covalent bond.

$$\text{Percentage of ionic character} = \frac{\text{Experimental value of dipole moment}}{\text{Theoretical value of dipole moment}} \times 100$$

Solved Example: Calculate the μ of HCl . If bond distance is 1.34 Å, charge = 4.8×10^{-10} esu, calculate % ionic character if experimental value of μ = 1.08 D ?

$$\mu = q \times d = 4.8 \times 10^{-10} \times 1.34 \times 10^{-8}\ \text{cm} = 6.432 \times 10^{-18}\ \text{esu cm.}$$

$$\% \text{ Ionic character} = \frac{1.08 \times 10^{-18}}{6.432 \times 10^{-18}} \times 100 = 16.79\ \%$$

- **Calculation of bond angle**

If μ_1 and μ_2 are bond moments and θ is the angle between their direction, then

$$\mu^2 = \mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 Cos\theta$$

In Water molecule there are two identical O-H bonds each bond moment od 1.5 debye. The net dipole moment of water molecule is 1.84 debye. Hence

$$1.84 = \sqrt{(1.5)^2 + (1.5)^2 + 2 \text{ x } 1.5 \text{ x } 1.5\ Cos\theta}$$

$$\theta = 104.5°$$

•Differentiating ortho, meta, and para isomers

In aromatic compounds, the para isomer typically has a zero dipole moment, while the ortho isomer usually has a higher dipole moment than the meta isomer.

Let us consider dichlorobenzene

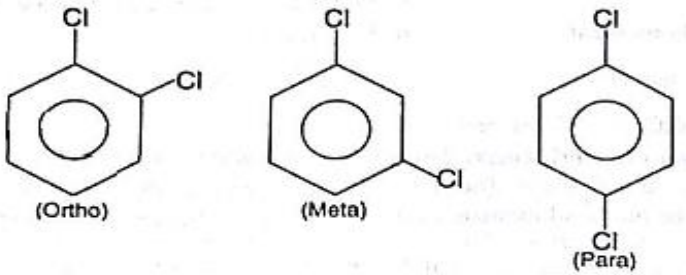


Fig. 8.10.

(*a*) *Ortho-isomer*

$$\mu = 2.6\ D$$

$$m_{C-Cl} = 1.5\ D$$

$$\therefore \quad \mu = \sqrt{m_1^2 + m_2^2 + 2m_1 m_2 \cos\theta}$$

$$2.6 = \sqrt{(1.5)^2 + (1.5)^2 + 2 \times (1.5)^2 \cos\theta}$$

$$\therefore \quad \theta = 60^\circ$$

Hence two C–Cl bond directions are *inclined* at angle of 60° to each other in ortho-isomer.

(*b*) *Meta-isomer*

$$\mu = 1.5\ D$$

$$1.5 = \sqrt{(1.5)^2 + (1.5)^2 + 2(1.5)^2 \cos\theta}$$

$$\therefore \quad \theta = 120^\circ$$

Hence in meta-isomer the angle between two C–Cl bond direction is 120°.

(*c*) *Para-isomer*

$$\mu = 0.0\ D$$

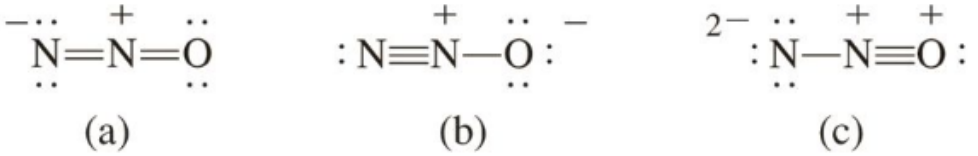
$$0.0 = \sqrt{(1.5)^2 + (1.5)^2 + 2(1.5)^2 \cos\theta}$$

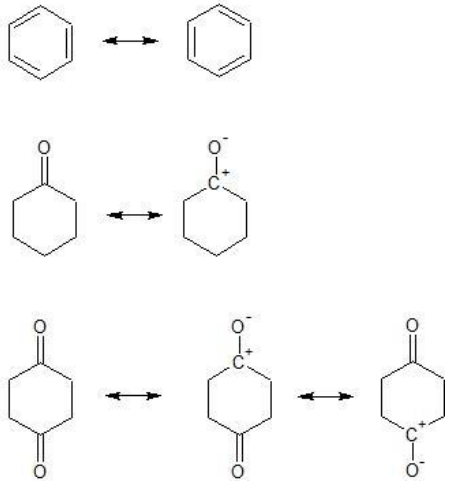
$$= 180^\circ$$

Hence both C–Cl bond directions are inclined at 180° to each other.

- **In deciding the resonance structure**

  - The study of dipole moment is of great help in determining the contribution of the possible structures in resonating system.





  - Have a high dipole moment.
  - Cannot have a linear structure.

Resonance can affect the polarity and dipole moment of a molecule, but only if at least one of the resonance structures involves formal charge separation, and even then there are cases where the overall molecule will not have a dipole moment.

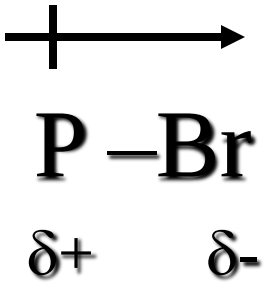
Resonance structures can be drawn for benzene, but none of them involve charge separation, therefore benzene is not polar and does not have a dipole moment. So just being able to draw resonance structures does not mean the molecule must be polar and have a dipole moment. Next let's examine cyclohexanone. In this case two resonance structures can be drawn and one of them involves charge separation. Therefor, we would expect this molecule to be polar and have a dipole moment. For 1,4-cyclohexanedione we can also draw resonance structures with formal charge separation, however in this case the symmetry of the molecule causes the local polarization at each carbonyl to cancel and overall, the molecule has no dipole moment.

Practice Drawing Dipoles

P- Br

P = 2.1

Br = 2.8



Practice

- H(2.1) – S(2.5)
- F(4.0) - C(2.5)
- C(2.5) - Si(1.8)
- N(3.0) – O(3.5)

Is it Polar?

- HF
- H_2O
- NH_3
- CF_4
- CO_2

$\mu = q \times r$

Percentage of ionic character $= \frac{\text{Experimental value of dipole moment}}{\text{Theoretical value of dipole moment}} \times 100$

$$P_M = \frac{\varepsilon - 1}{\varepsilon + 2} \cdot \frac{M}{\rho} = \frac{4}{3} \pi N \alpha_D$$

$\mu^2 = \mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\theta$

$$P_M = \frac{\varepsilon - 1}{\varepsilon + 2} \cdot \frac{M}{\rho} = \frac{4\pi N}{3} \left(\alpha_D + \frac{\mu^2}{3kT} \right)$$

$\mu = 0.0128 \times 10^{-18} \sqrt{B}$ esu-cm

$\mu = 0.0128 \times \sqrt{B}$ debey

$\mu = \sqrt{9kT/4\pi N (P_{gas} - P_{solid})}$

$\mu = 0.0128 \sqrt{[(P - P_i)T]}$

$P = (n^2-1)/(n^2+2) \cdot M/d$

Prob. 1:

For $SO_2(g)$, permanent dipole moment is 1.63D, dielectric constant is 1.000993 and density is 2.6288 kg/m^3 at room temperature. Calculate (a) the total polarization, (b) orientation polarization and (c) polarizability.

Soln.

$P_t = (\varepsilon-1)/(\varepsilon+2)\times M/\rho = 4\pi N\alpha_D/3 + 4\pi N/3(\mu^2/3kT)$

Here, $\varepsilon = 1.00993$

$\mu = 1.63D = 1.63\text{x}10^{-18}$ esu cm

M = 64.06 g mol-1

$\rho = 2.6288$ kg/m^3 $= 2.6288\text{x}10^{-3}$ g cc^{-1}

$k = 1.38\text{x}10^{-16}$ erg deg^{-1}

$N = 6.023\text{x}10^{23}$

T = 298 K

(a)

$P_t = (1.009931)/(1.00993+2)\times 64.06/2.6288\times 10^{-3}$

$= 80.39$ cc mol^{-1}

(b)

$P_o = 4\pi N/3(\mu^2/3kT)$

$= 4\times 3.14\times 6.023\text{x}10^{23}\times(1.63\text{x}10^{-18})^2/\ (9\times 298\times 1.38\text{x}10^{-16})$

$= 54.3$ cc mol^{-1}

(c)

$P_t = P_o + 4\pi N\alpha_D/3$

Or, $\alpha_D = 3\times(P_t - P_o)/4\pi N$

$= 3\times(80.39 - 54.3)/(4\times 3.14\times 6.023\times 10^{23})$

$= 1.03\times 10^{-23}$ cc

Prob. 2:

At 0^0C, the molar polarization of a liquid (M = 85 g mol^{-1}) is 32.16 cc mol^{-1} and its density is 1.92 g cc^{-1}. Calculate (a) the dielectric constant value, and (b) polarizability of that compound.

Soln.

We know

$$P_m = (\varepsilon-1)/(\varepsilon+2)\times M/\rho = 4\pi N\alpha_D/3$$

Here,

P_m = 32.16 cc mol^{-1}

ρ = 1.92 g cc^{-1}

M = 85.0 g mol^{-1}

ε = ?

and α_D = ?

N = 6.023×10^{23}

(a)

$$P_m = (\varepsilon-1)/(\varepsilon+2)\times M/\rho$$

$$(\varepsilon-1)/(\varepsilon+2) = P_m\times\rho/M = 32.16\times1.92/85$$

$$= 0.73$$

0r, ε = 9.11

(b)

$$P_m = P_i = 4\pi N\alpha_D/3$$

Or, $\alpha_D = 3\times P_m/4\pi N$

$= 3\times32.16/(4\times3.14\times6.023\times10^{23})$

$= 1.27\times10^{-23}$ cc

Prob. 3:

The dipole moment of LiF in the gaseous state is 6.67D, the intermolecular distance is 0.152 nm. Calculate the ionic character of LiF.

Soln.

We know

$$\mu = q \times d$$
$$= 4.8 \times 10^{-10} \times 0.152 \times 10^{-9}$$
$$= 7.3 \times 10^{-18} \text{ esu cm}$$
$$= 7.3\text{D}$$

Therefore,

$$\% \text{ of ionic character} = (\mu_{obs}/\mu_{cal}) \times 100$$
$$= (6.67/7.3) \times 100$$
$$= 92\%$$

Prob. 4:
Calculate the dipole moment of o-dichlorobenzene and o-dinitrobenzene, if μ_{C-Cl} = 1.59D and μ_{C-NO2} = 4.01D.

Soln.

For disubstituted benzene, we know

$$\mu_t^2 = \mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta$$

Here,

μ_t = total dipole moment

μ_1 and μ_2 are the bon moment

θ is the angle between the bonds

For o-dichlorobenzene and o-dinitrobenzene, $\mu_1 = \mu_2$

For o-dichlorobenzene, μ = 1.59D, for o-dinitrobenzene, μ = 4.01D and $\theta = 60^0$

For o-dichlorobenzene

$\mu_t^2 = 2\mu^2 + 2\mu^2\cos\theta$

or, $\mu_t = \sqrt{(2\mu^2 + 2\mu^2\cos\theta)}$

$= \sqrt{[2\times(1.59)^2 + 2\times(1.59)^2\cos60]}$

$= \sqrt{[7.58]}$

= 2.75D

For o-dinitrobenzene

$\mu_t = \sqrt{(2\mu^2 + 2\mu^2\cos\theta)}$

$= \sqrt{[2\times(4.01)^2 + 2\times(4.01)^2\cos60]}$

$= \sqrt{[48.24]}$

= 6.94D